VANIER COLLEGE

Faculty of Science and Technology
Computer Science Department

Advanced Networks (420-521-VA) - Fall 26

Lab 7: Planning your Team Project

VPNs

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Overview

A VPN (Virtual Private Network) is a tool that creates a secure, encrypted connection between a device and the internet.

When a VPN is used...

- **Remote Server**: Internet traffic is sent through a VPN server operated by the VPN host
- **IP Address Masking**: real IP address is hidden and replaced with the VPN server's IP address
- **NAT Traversal**: allows VPN traffic to pass through NAT devices and routers without breaking the connection.
- **Traffic Routing**: Data is routed through the VPN server before reaching online services
- **VPN tunnel / Tunnelling**: an encrypted connection between the device and the remote VPN server over the internet to protect data transmission.
- **Encryption Keys**: encrypt and decrypt data to keep communication between devices secure.

VPN is for:

- Online Privacy
- Online Identity Protection
- Network Security
- Geolocation Hiding
- Third-Party Prevention

In short, VPNs are used wherever security and privacy are needed.

VPN doesn’t hide:

- **User digital fingerprint** (it contains information like device and browser settings)
- **User identity**: If you sign in to any personal account or make a debit or credit card payment while using a VPN, your identity will be revealed.
- **Online threats**: you're still exposed to malicious content. A VPN is only as secure and as smart as the person using it. If you download questionable files or visit suspicious websites while using a VPN, you are still at risk of being attacked.

Software & Hardware

Hardwares:

- 1 Physical Computer
- 2 virtual Linux machines

Softwares:

- Linux Kernel Networking (ip netns)
- WireGuard: VPN software tool for establishing a secure tunnel.
- Tcpdump or Wireshark for packet capture
- Tailscale for Commercial/Mesh Overlay Service
- RiseupVPN, Calyx VPN or Autistici/Inventati for Community-run Privacy service

Implementation Plan

We will implement this project by running a single lightweight Linux virtual machine. Inside this Linux VM, using the kernels built in Network Namespaces (ip netns), we’ll create two isolated virtual spaces (one for the Client and one for the Server). To connect them together, we’ll build a WireGuard tunnel between those two namespaces, route traffic, and run tcpdump to capture the packets.

This approach not only keeps our physical Linux computer safe from any potential issues we might have, but it’ll also run faster than having two heavy virtual machines. This plan follows a container-like approach.

Research References

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